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Integrated Characterization and Reservoir Potential of the Heavy Mineral Deposits Hosted by Carbonate Sediments in Saudi Arabia

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The carbonate sediments of the study area represent a major sediment fill that covers an area of around 3.7 million square kilometers across the Arabian Gulf region, hosting one of the most prolific non-associated gas plays in the world (Alsharhan, 2006; Pöppelreiter and Marshall, 2014). The shallow marine carbonate sediments of the study area were deposited near the equator due to a major transgression on top of thick siliciclastic deposits coinciding with the icehouse-greenhouse transition (Fig. 1). The hydrocarbon play is part of the Paleozoic petroleum system sourced primarily from the organic-rich Silurian Qusaiba shales, which are considered the source rocks for 80-90% of all Paleozoic hydrocarbons in many prominent oil and gas fields in North Africa and Arabia (Mahmoud et al., 1992; Nairn and Alsharhan, 1997). The seals of the study area include different types: impermeable bedded anhydrite layers, tight carbonate layers, and the thick shale layers of the overlying formation, which is considered a regional seal that separates this Paleozoic petroleum system from the Mesozoic petroleum system (Konert et al., 2001, Alqahtani, 2019).

The formation sediments comprise carbonates and evaporites (from highest to lowest), dolomite, limestone, and anhydrite. Traces of auxiliary minerals: celestite, pyrite, barite, chert, fluorite, quartz, and clay minerals are present in the formation with no effect on reservoir quality. The study area includes three separate units two from the Late Permian and one from the Early Triassic, confined within a salt-cored structure resulting from the Precambrian Hormuz salt. This near-circular structure is associated with prominent deeply rooted faults that are detectable from seismic data, acting as pathways to hydrocarbons and other metal-rich hydrothermal fluids (Fig. 2) (Stewart, 2018; Alqahtani, 2019). The lithology mainly consists of clean carbonates and evaporites, consisting of dolomite, limestone, and anhydrite. Reservoir quality is controlled by the interplay between original depositional factors and post-depositional (diagenetic) factors. Dolomitization, dissolution, and anhydrite cementation are the most critical diagenetic processes, impacting reservoir quality positively and negatively. The study area has eight depositional lithofacies, and they are named after their depositional environments. The lithofacies (from updip to downdip) are subaerial exposure, sabkha/saline anhydrite, tidal flat, lagoon, shallow subtidal, shoal flank, shoal crest, and open marine. The lithofacies succession represents deposits of shallow marine homoclinal ramp within a tectonically stable epeiric sea. Based on the level of restriction, lithofacies can be grouped into three depositional zones: marginal marine, barrier, and normal marine (Fig. 3). The marginal marine includes five back-barrier lithofacies. This zone shows significant restriction in the form of muddy texture, rare

restricted biota, and extensive dolomite and anhydrite lithologies. The barrier zone includes the grain-dominated shoal crest and shoal flank, which consist primarily of ooids, peloids, bioclasts, and intraclasts. Grain-dominated shoal complex lithofacies are reservoir rocks and the most diagenetically prone facies where dolomitization, dissolution, calcite, and anhydrite cementations occur. Molds are the most common pore type when dissolution dissolves metastable aragonitic ooids and fossils. Dolomitization connects molds by the porous crystalline matrix that is also prone to fracturing. Anhydrite cementation postdates dolomitization and kills all pore types. The normal marine zone includes deeper water open marine lime mudstone lithofacies, the least diagenetically altered lithofacies with open marine fossils (Alqahtani, 2019). This core-based study aims to identify types of heavy mineralization and examine their effect on well logs and the pore system within the mineralized intervals.

Introduction

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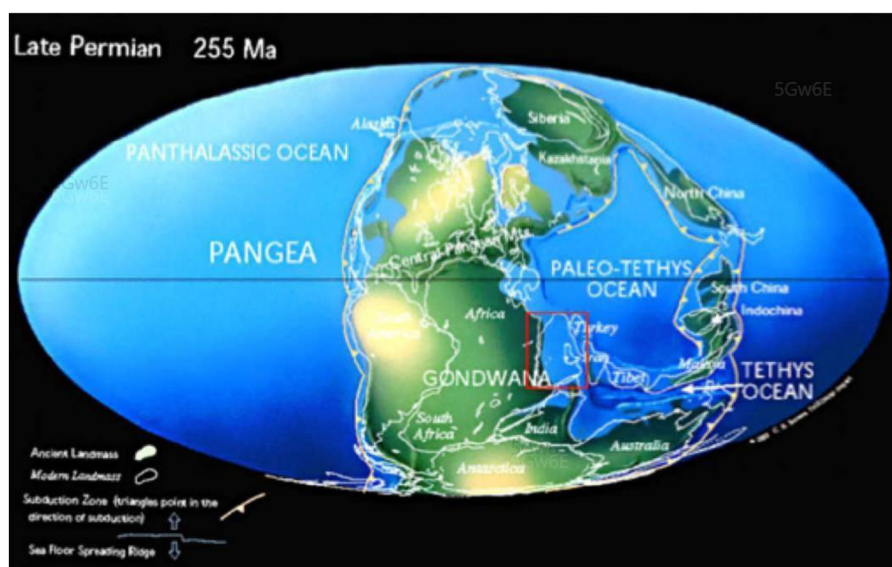


Figure 1—Paleogeographic map showing the position of the Arabian Plate (red rectangle) during the Late Permian time. Modified from Scotese, 2002.

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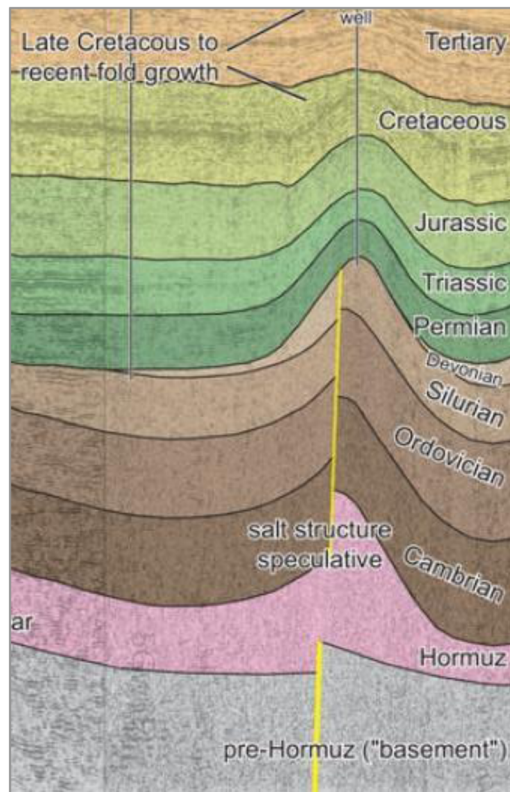
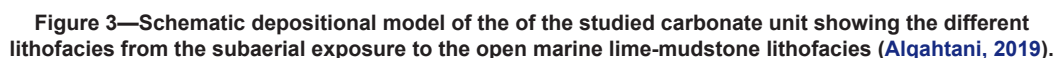


Figure 2—Interpreted seismic section of the study area (salt-cored structure) showing the Precambrian basement, Hormuz salt, the overlying Phanerozoic cover rocks, and the deeply rooted fault system, which is considered a pathway for hydrocarbons and metal-rich hydrothermal fluids. (Modified from Stewart, 2018).



The study is based on data from a single cored well with over 400 feet of core across the studied succession. This integrated study utilizes four methods: (1) core description covering the carbonates and the mineralized intervals recording semi-quantitative lithology, organic and inorganic grain types, grain sizes, the modified Dunham's classification, and depositional cycles; (2) petrophysical data includes wireline logs (spectral gamma-ray, bulk density, neutron porosity, resistivity, photoelectric factor, and caliber), routine core plug analyses (porosity, permeability, and grain density), and core-gamma-ray logs to define heavy mineral logging criteria and their reservoir potential; (3) petrographic description of Alizarin Red-S stained thin sections to identify semi-quantitative mineral types (including heavy minerals), grain types, depositional facies, diagenetic facies, semiquantitative pore types, texture, dolomite types, anhydrite types, dolomite crystal sizes, calcite cement types, dissolution, and a simple paragenetic sequence of diagenetic processes and their cross-cutting relationships. The petrographic description enabled the selection of several thin sections to conduct scanning electron microscopy (SEM) and energy dispersive X ray spectroscopy (EDS) analyses to map minerals; (4) XRD data from selected plugs to geochemically analyze carbonate, evaporite, clay and more importantly heavy minerals that have been recorded from core and petrographic descriptions. All well data were loaded into the Techlog platform and the core gamma ray were used to depth shift core data with wireline logs.

The core description revealed typical Late Permian deposits similar to other wells in the field, but this time revealed one-of-a-kind, thick, fully mineralized horizons hosted within dolostone and anhydrite layers. This mineralization is the Mississippi Valley Type (MVT) deposits that occur in sedimentary carbonate rocks typically in a passive-margin setting and found in different parts of the world (Fig 4). The heavy metals were carried by oxidized hydrothermal brines, through faults and permeable carbonate deposits, from basement rocks to be dropped out in reduced environments (Robb, 2005; Tylor et al., 2009; Leach et al., 2010a). The fully mineralized core specimens are heavier than normal showing brown sphalerite (ZnS), metallic galena (PbS), brassy pyrite (FeS₂) and barite (BaSO₄) occurring in different concentrations, as later confirmed by the XRD analysis. Unlike host rocks, vuggy porosity dominates pore types and is clearly visible with a

naked eye. MVT layers show no signs of carbonates or anhydrites, and they are tied really well with logs; density in particular. The integrated wireline logs with core descriptions, and routine core analysis revealed a total of nine high-density horizons where density reading exceeds 3.1 g/cc for the density of bedded anhydrite averaging of more than 3.1 g/cc (Fig. 5). Thus, core descriptions show three non-carbonate zones of heavy minerals of sphalerite, galena and pyrite. There are three types of heavy minerals identified in the cored section: 1- non-carbonate, replacive mineralized zones (except for minor late calcite cementation) that show a distinct hyper- density behavior, high PEF log, and very low resistivity reading; 2- dolomitized-grainstone-associated disseminated heavy minerals (up to 18% heavy minerals); 3- dolomitized-grainstone-associated barite cement (up to 35%) (Fig. 5). The last two types show high density, PEF, and normal resistivity readings. Mineralized zones lack carbonates and are associated with dissolution, vuggy, and fracture porosity, which enhance their pore system potential. The disseminated, replacive heavy mineral deposits do not significantly influence reservoir quality, whereas barite cementation fills the interparticle porosity, diminishing pore throats and therefore connectivity. Most heavy mineral deposits are fault-related and accompanied by dissolution, pore filling, and replacement of host carbonate rocks. Saddle dolomite and late calcite cementation are associated with the heavy minerals and barite, indicating hydrothermal- induced late diagenetic processes (. The carbonate-hosted heavy minerals are associated with high- temperature brines carrying metals from basement rock through deep-rooted faults (detected by seismic) near the cored well.



Figure 4—Districts and deposits distribution of the Mississippi Valley Type deposits, Jabal Dhaylan by the Red Sea was the only known place for MVT in the Arabian Peninsula (Leach et al., 2010b).

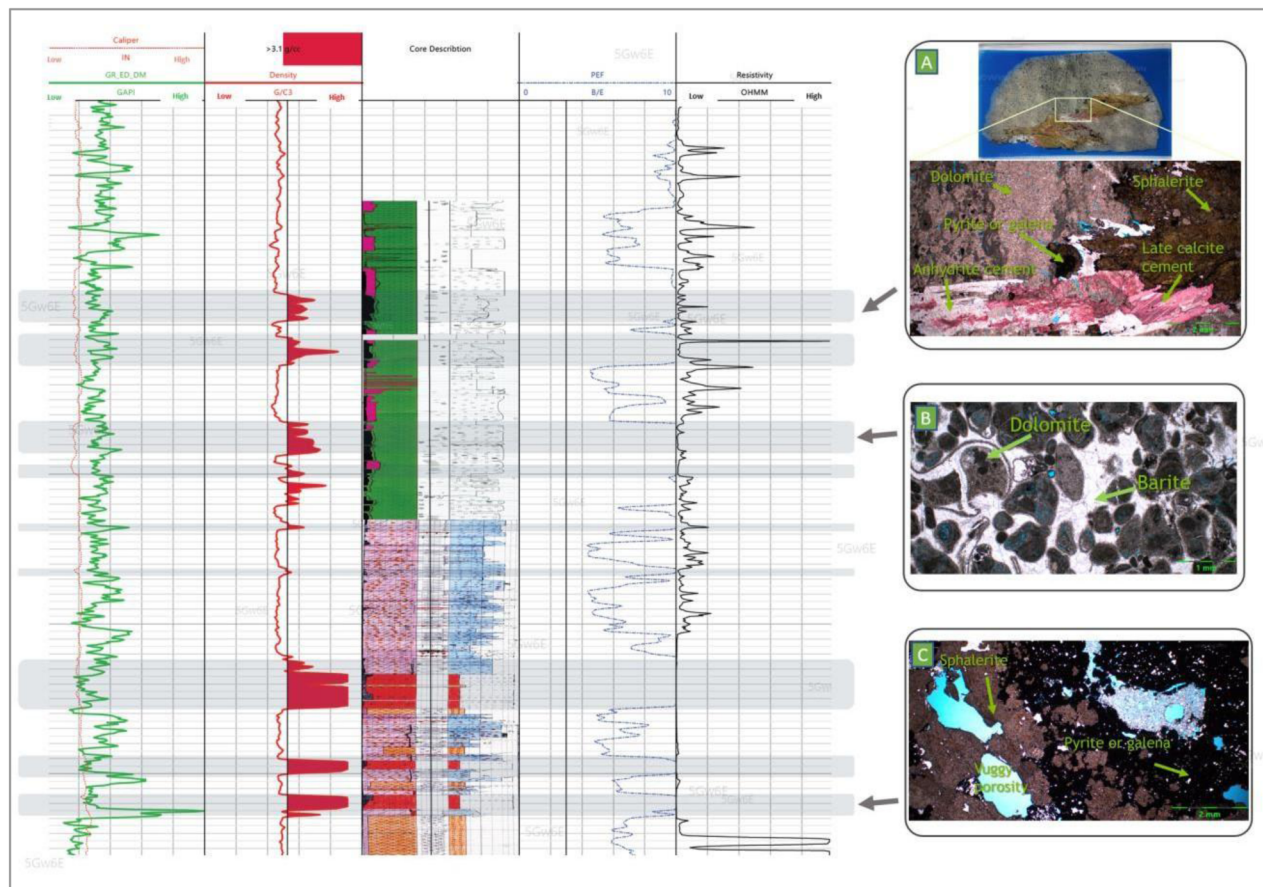


Figure 5—Integrated log display that includes wireline logs and depth-shifted core descriptions showing the nine high-density intervals within each of which the density logs read more than 3.1 g/cc and the PEF reads high around 10. The lower three horizons represent the thick, fully mineralized non-carbonate MVT sphalerite, galena, and pyrite deposits.

All of which have an enhanced pore system. A- Thin section image and zoomed in photomicrograph showing the disseminated sphalerite hosted by dolomitized peloidal packstone; the late-stage coarse-crystalline calcite cementation is associated with anhydrite cement. **B-** photomicrograph showing barite cement (another dense, hydrothermal mineral), filling the intergranular pores between microporous dolomitized grains. A and B cannot be seen from the core description. Following-up petrography and geochemistry were crucial to characterizing and quantifying those diagenetic features and explaining the ultra-high density readings. **C-** Photomicrograph of the non-carbonate fully mineralized MVT deposits (sphalerite, pyrite, and galena), showing vuggy, intercrystalline and micro pore types; pyrite and galena look black under the microscope, which requires further EDS analysis to determine their chemical compositions.

The XRD analysis revealed the occurrence of MVT minerals, showing that sphalerite ranges from 1 to 75% with an average reading of 40%; galena ranges from 1 to 51% with an average reading of 11%; pyrite is the most common sulfide, ranging from 1 to 87% with an average reading of 37%. The mineralized sections are non-carbonates and are associated with visible vuggy porosity and rare off-white, coarse-crystalline saddle dolomite. The visible vuggy porosity has not been recorded in the host rocks which involves different pore types ranked (from common to rare): intercrystalline, moldic, interparticle, fracture, micropores, and vuggy. Petrography of mineralized thin sections shows only three types: vuggy, intercrystalline and fracture porosities. Sphalerite is characterized by massive brown crystals, whereas the pyrite and galena are dark-colored crystals (Fig. 6). In general, MVT minerals occur in the fully mineralized zones in a replacive form showing no signs of carbonates or anhydrite. Other minerals associated with the MVT sulfides are saddle dolomite, barite, and late calcite cementation; all are present as cements filling voids. Barite cements are present up to 5% in the fully mineralized zones, but in the shallower horizons, they can reach up to 35%. Saddle dolomite crystals show clear sweeping extension under cross-polarized light. This study's results prove the occurrence of MVT minerals in the Late Permian rocks as good reservoir quality and high-grade ores.

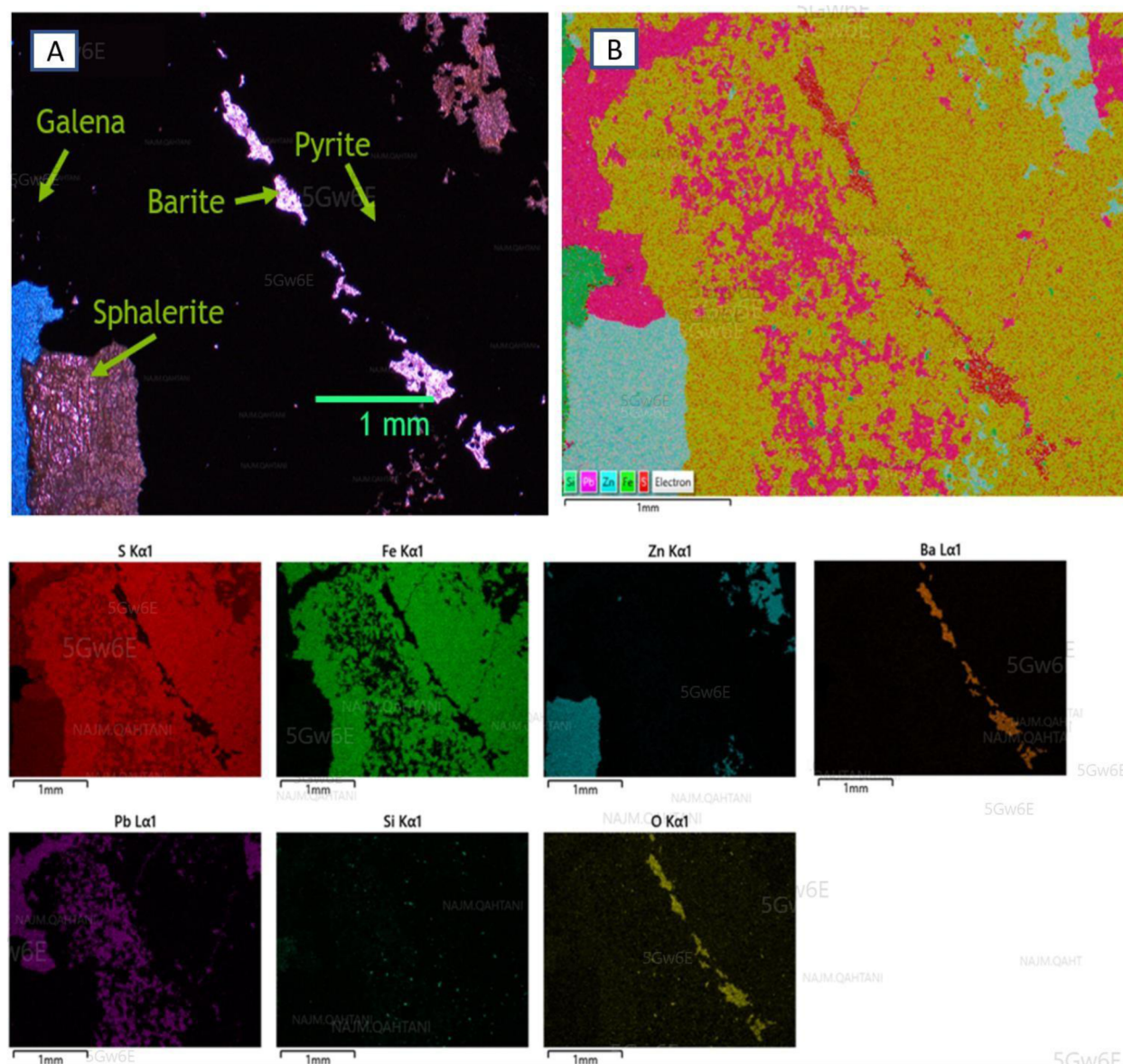


Figure 6—A- Photomicrograph of MVT mineralized thin section showing the four heavy minerals (sphalerite, galena, pyrite, and barite); pyrite and galena are dark-colored and challenging to tell apart. **B-** EDS image of color-coded minerals; pyrite (gold) dominates mineralogy. The elemental mapping of each element is illustrated in the smaller seven images. These images prove the need to integrate results using different types of analyses.

Discussion and conclusions

Most of the carbonate rocks found in the Arabian platform subsurface and outcropping fall within the evaporitic calcite dolomite with minor quantities of clays and wind-blown quartz grains. The MVT lead-zinc deposits occur in different parts of the world, and are mainly hosted by carbonates (dolomite, siderite, ankerite, and calcite). Serendipity plays a prominent role in discovering the mineralized deposits within the Late Permian carbonate deposits in this study, anchoring for the first time high- grade MVT deposits that are characterized by core descriptions, logs, petrography, and XRD analyses as: 1- non-carbonate fully-mineralized sections with good reservoir quality that have distinct density behavior of >4.6 g/cc, high PEF log, and very low resistivity reading; 2- disseminated MVT minerals within dolomitized grainstone facies (MVT up to 18%), and this type shows random elevated density readings of more than 3.1 g/cc, high PEF, and normal resistivity readings showing no impact on reservoir quality; 3- Intergranular-filling barite cements filling the void space between dolomitized grains, showing random elevated density readings of

more than 3.1 g/cc, high PEF, and normal resistivity readings and negative impact on the pore system. Even with small amounts, the disseminated replacive MVT deposits and barite cement significantly increase the density logs.

In contrast to the hosted carbonate rocks, the fully mineralized cored sections are heavy and metallic, and do not react with applied acid. Their associated logs show ultrahigh density (more than 4.5 g/cc), PEF (around 10), and low resistivity measurements due to the high conductivity of the metallic deposits. The recorded sulfide minerals are a mix of pyrite, galena, and sphalerite, and the associated pore system is different, dominated by vuggy porosity that can reach up to macropores in scale. Since the mineralized zones are porous and permeable, they may represent upside potential for hydrocarbons, positively impacting production and increasing reserves due to vuggy and the associated intercrystallite pores. On the other hand, barite cement, which is associated with high density (between 3.1 and 4.5 g/cc) and PEF (around 10) readings, fills vugs and fractures in the fully mineralized intervals and obliterates intergranular porosity in the younger high-density dolomitized intervals, isolating microporous ooids. The disseminated MVT minerals within the carbonate rocks have little to no impact on reservoir quality and are associated with high density (between 3.1 and 4.5 g/cc) and high PEF log readings (around 10). Resistivity logs read normal for the barite-cemented and the disseminated heavy minerals carbonate intervals.

The logging criteria generated from this study can be part of an early-stage exploration program leveraging on oil and gas well data for MVT minerals and other associated rare elements. It would start by log scanning to define high-potential MVT deposits showing high density (more than 3.1 g/cc), followed by geochemical sampling from the flagged wells to assess elemental compositions by XRD, XRF, ICP-MS, and ICP-OES. High-resolution elemental analysis, like ICP, is needed to define MVT deposits and their associated trace elements since sphalerite is the primary source of Energy Critical Elements (ECEs). The ECEs are critical in manufacturing solar panels, semiconductors, wireless infrastructure, and power electronics, including indium, gallium, and germanium. ECEs are not found in nature as individual mineral deposits, but they form as trace elements of MVT base metal deposits; for example, around 95% of the recovered indium globally is produced from zinc ore processing (Lokanc et al., 2015).

After examining the MVT mineral system components that led to such a discovery, we discovered that most of the MVT ingredients are associated with the study area. The studied field is a four-way closure, salt-centered structure, associated with deep-rooted faults. The study area shows a near-circular negative gravity anomaly indicating a salt-cored structure, provided that the Hormuz salt occurs at a 1000-km depth far beyond the reach of any drilled wells (Leach et al., 2005; Edgell, 1996; Stewart, 2018).

The carbonate-hosted MVT heavy minerals are associated with high-temperature brines carrying metal-rich brines from basement rocks through deep-rooted faults that radiate around the salt-induced structure. Faults are conduits not only for hydrocarbons but also for the later-stage dissolution and mineralization fluids. The sedimentological, petrographic, and XRD results show that sphalerite and galena were recorded in high grades exceeding the combined lead and zinc ore grades of known MVT deposits globally (Christie and Brathwaite, 1995; Paradis et al., 2007; Leach et al., 2010b). The economic potential of such an MVT mine ore may be challenged by its depth and location.

The integrated results of this core-based study, which include core descriptions, petrographic, SEM, and chemical analyses, can be utilized by petrophysicists to better understand and reprocess wireline logs for all wells that show high-density low-resistivity readings. Mineralized zones with vuggy, intercrystalline, and fracture porosities represent reservoir potential that can be considered in reserve calculations.

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